GridSignal Mini

Technical Evaluation Submission Report & Architecture Dossier

Candidate: Amit Kumar	**Evaluator:** Sariya Khan (sariyakh25@gmail.com)
Tech Stack: Python 3.13 · FastAPI · SQLite · React 18	**Test Suite: 16 / 16 Passed (1.88s)**
Live Vercel URL: https://grid-signal-mini.vercel.app/	**GitHub Repo:** https://github.com/amitkumar824/GridSignal-Mini

1. Deliverables Checklist & Compliance

Required Deliverable	Location / URL	Status
1. Vercel Live Deployment	https://grid-signal-mini.vercel.app/	■ **Verified & Deployed**
2. GitHub Repository	https://github.com/amitkumar824/GridSignal-Mini	■ **Complete (main)**
3. Dashboard Screenshot	docs/dashboard_screenshot.png (Embedded below)	■ **Captured Live**
4. Architecture Diagram	architecture.md & README.md	■ **Documented**
5. Automated Test Suite	16 tests in tests/test_app.py	■ **16 / 16 Passed (1.88s)**
6. Local Run Instructions	README.md (1-command run)	■ **Tested & Working**

2. Live Vercel Dashboard Screenshot

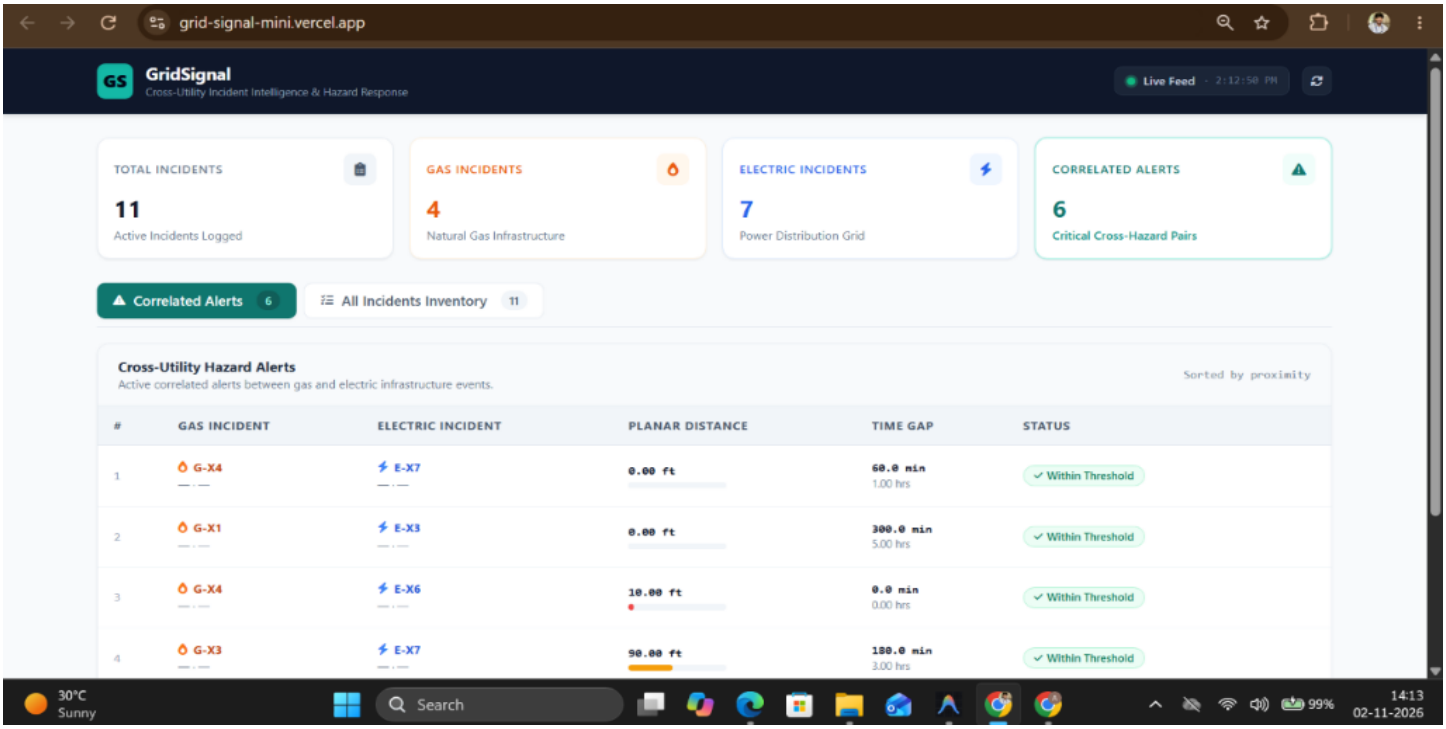


Figure 1: Live GridSignal Mini Dashboard on Vercel displaying 11 active incidents, 6 proximity-correlated hazard pairs, and auto-refreshing telemetry.

3. System Architecture & Data Flow

GridSignal Mini implements a modular, transport-agnostic pipeline. Incident data enters via either batch CSV files (CLI) or real-time JSON webhooks (POST /incidents). The **Ingestion Engine** enforces deterministic Last-Write-Wins (LWW) deduplication, corrupt ISO-8601 timestamp filtering, and stores records in SQLite with WAL mode concurrency. The **Correlation Engine** computes Euclidean spatial planar proximity (≤ 200 ft) and temporal coexistence (≤ 6 hours) between independent GAS and ELECTRIC utility events, emitting sorted hazard alerts to the React SPA.

4. Mathematical Formulation & Conflict Policy

Cross-Type Correlation Conditions (Both Inclusive):

- **Planar Distance:** $\Delta d = \sqrt{(x1 - x2)^2 + (y1 - y2)^2} \leq 200.0$ ft
- **Temporal Window:** $\Delta t = |t_gas - t_electric| \leq 360.0$ minutes (6.0 hours)
- **Missing Coordinates:** Incidents with NULL coordinates are safely stored but strictly excluded from matching.

Deterministic Deduplication & Last-Write-Wins (LWW):

- **Identical Duplicates:** Skipped with 0 redundant database writes.
- **Conflicting Records:** Updated in-place based on incident_id (LWW semantics).
- **Malformed Timestamps:** Logged with warnings and skipped without crashing batch ingestion.

5. Automated Test Suite (16 / 16 Tests Passed)

```
tests/test_app.py::TestBoundaryAndCorrelation::test_exact_boundary_conditions PASSED [ 6%]
tests/test_app.py::TestBoundaryAndCorrelation::test_boundary_distance_just_outside PASSED [12%]
tests/test_app.py::TestBoundaryAndCorrelation::test_boundary_time_just_outside PASSED [18%]
tests/test_app.py::TestBoundaryAndCorrelation::test_missing_coordinates_never_matches PASSED [25%]
tests/test_app.py::TestBoundaryAndCorrelation::test_time_ordering_reversibility PASSED [31%]
tests/test_app.py::TestBoundaryAndCorrelation::test_correlation_ignores_severity_status_area PASSED [37%]
tests/test_app.py::TestBoundaryAndCorrelation::test_one_to_many_matches PASSED [43%]
tests/test_app.py::TestIngestionAndDeduplication::test_idempotent_ingestion PASSED [50%]
tests/test_app.py::TestIngestionAndDeduplication::test_conflicting_duplicates_last_write_wins PASSED [56%]
tests/test_app.py::TestIngestionAndDeduplication::test_malformed_timestamp_gracefully_skipped PASSED [62%]
tests/test_app.py::TestIngestionAndDeduplication::test_missing_coordinates_stored_as_null PASSED [68%]
tests/test_app.py::TestApiEndpoints::test_get_health PASSED [75%]
tests/test_app.py::TestApiEndpoints::test_get_incidents PASSED [81%]
tests/test_app.py::TestApiEndpoints::test_get_alerts PASSED [87%]
tests/test_app.py::TestApiEndpoints::test_get_summary PASSED [93%]
tests/test_app.py::TestApiEndpoints::test_create_incident_json PASSED [100%]
Result: 16 passed in 1.88s
```

6. Quick Start & Local Execution

